

PHASE 2: REQUIREMENT ANALYSIS

Core Project Objectives:

To ingest custom user parameters, parse abstract plots into strict 5-panel layout matrices, generate context-aware conversational character dialogues, download graphic illustrations, and compile files into a clean offline PDF document format safely.

Functional Requirements Array:

Allow parameter collection (story prompt, main character name, setting, tone, art style preference).

Validate and process user payloads via asynchronous backend application routes.

Call Gemini API models for structural blueprint parsing and script dialogue generation.

Trigger automated diffusion nodes for panel-wise JPEG asset caching on hard disk slots.

Compile asset rows into downloadable print-ready vector documents with timestamped names.

Non-Functional Requirements Boundary:

Deliver complete multi-panel visual comic publications in under 120 seconds, ensure intuitive user navigation grids, isolate private API keys securely inside .env configurations, and utilize robust try-except error containment blocks to defend against microservices failure boundaries.

Hardware Allocation Baseline:

Minimum Quad-core CPU configuration running at >2.5 GHz clock speed, 8 Gigabytes physical memory capacity (RAM), 500 MB available solid-state storage disk space, and active broadband internet infrastructure connection

Software Configuration Environment:

Modern web browser instances (Chrome/Edge/Firefox), Visual Studio Code Core IDE workspace, Python Development Kit SDK 3.10.x/3.11.x runtime setups, and GitHub repository hosting framework platform.